Emre Ertürk

emreerturk@std.iyte.edu.tr · emreerturk4585@gmail.com · github.com/EmreErturk-3 · emreerturk-3.github.io

RESEARCH INTERESTS

Evolutionary computation, artificial life, swarm robotics, evolutionary and developmental robotics, bio-inspired robotics, robot locomotion.

EDUCATION

B.Sc. in Computer Engineering, İzmir Institute of Technology 2022–present

CGPA 3.12 / 4.00 (3.83 in fall 2025–26). Thesis: Smart Zebra, advised by Assoc. Prof. Selma Tekir, funded by TÜBİTAK 2209-A.

Relevant courses: Behavioral Robotics (AA), Evolutionary Computation (BA), Embedded Computer Systems (BA), Analysis and Design of Algorithms (BA), Differential Equations (AA).

İzmir Atatürk Lisesi High school

PAPERS AND REPORTS

O. Kansoy, **E. Ertürk** (equal contribution). Smart Zebra: A Prompt-to-GRPO Recipe for Logic-Grid Reasoning in Small Language Models. B.Sc. thesis, manuscript in revision, 2026.

E. Ertürk. AEASS v3: Revisiting an Adaptive Evolutionary Algorithm for Solving Sudokus. Technical report, 2026.

E. Ertürk. Slavemaker Swarm: An Individual-Based Simulation of Polyergus Slave Raids on Formica Colonies. Project report, 2026.

H. N. Kömürcü, D. Karabacak, **E. Ertürk**, T. İyiarı. An Indoor Mobile Robot (P.C.B) for Corridor Navigation and Secure Delivery Using IMU-Aided Control and ArUco Landmarks. CENG 483 course paper, 2026.

PROJECTS

EvoMaxx, artificial life sandbox on the GPU 2026–present

WebGL2. Controlled experiments on about 18,000 worlds; raised the share of lifelike genomes from 7% to 95%. Reproduced twelve papers it builds on.

Slavemaker Swarm, agent-based model of ant slave raids 2026

3-D simulation with pheromone fields, colony takeover and coevolution.

AEASS v3, memetic algorithm for Sudoku 2025–2026

GA with local search; all 65,201 runs solved over 52,458 benchmark puzzles. Python, C, WebGL2.

P.C.B, indoor delivery robot (CENG 483, team of four) Fall 2025

Arduino Mega and Jetson Nano; IMU-aided PID heading control; ultrasonic navigation; ArUco landmarks.

PartyBot, gesture-controlled mecanum robot (CENG 424, team of four) Spring 2026

Two ESP32 boards over ESP-NOW.

SKILLS

Python, C, JavaScript/TypeScript, GLSL, WebGL2, ARGoS, Arduino, ESP32, Jetson Nano, OpenCV.

LANGUAGES

Turkish (native), English (C1, YDS 93.5), Chinese (elementary), German (beginner). ALES 85.